CENG381 Stochastic Processes

CTMC and Birth-Death Processes - Answer Key 1

Q1 Solution

a) Total leaving rates and holding times:

$q_0 = -Q(0,0) = 1/3 \Rightarrow H_0 \sim \text{Exponential}(1/3)$, $E[H_0] = 3$ time units

$q_1 = -Q(1,1) = 1/2 \Rightarrow H_1 \sim \text{Exponential}(1/2)$, $E[H_1] = 2$ time units

$q_2 = -Q(2,2) = 1/2 \Rightarrow H_2 \sim \text{Exponential}(1/2)$, $E[H_2] = 2$ time units

States 1 and 2 have the shortest expected holding time (2 time units each). State 0 (Off) stays off the longest on average (3 time units).

b) Global balance equations (pi*Q = 0):

For each state i: (rate leaving i) = (rate entering i)

State 0: $\pi(0)\cdot(1/3) = \pi(1)\cdot(1/3) + \pi(2)\cdot(1/6)$

State 1: $\pi(1)\cdot(1/2) = \pi(0)\cdot(1/4) + \pi(2)\cdot(1/3)$

State 2: $\pi(2)\cdot(1/2) = \pi(0)\cdot(1/12) + \pi(1)\cdot(1/6)$

c) Solving for stationary distribution:

From the State 0 equation (multiply both sides by 12):

$4\pi(0) = 4\pi(1) + 2\pi(2)$

$\Rightarrow 2\pi(0) = 2\pi(1) + \pi(2)$ ... (i)

From the State 1 equation (multiply both sides by 6):

$3\pi(1) = (3/2)\pi(0) + 2\pi(2)$

Multiply by 2: $6\pi(1) = 3\pi(0) + 4\pi(2)$ ... (ii)

From normalization: $\pi(0) + \pi(1) + \pi(2) = 1$ ... (iii)

From (i): $\pi(2) = 2\pi(0) - 2\pi(1)$

Substitute into (ii):

$6\pi(1) = 3\pi(0) + 4(2\pi(0) - 2\pi(1))$

$6\pi(1) = 3\pi(0) + 8\pi(0) - 8\pi(1)$

$14\pi(1) = 11\pi(0)$

$\pi(1) = 11/14 \cdot \pi(0)$

Then: $\pi(2) = 2\pi(0) - 2(11/14)\pi(0) = 2\pi(0)(1 - 11/14)$

$= 2\pi(0)(3/14) = 3/7 \cdot \pi(0)$

From (iii): pi(0) + (11/14)*pi(0) + (3/7)*pi(0) = 1

pi(0) * [1 + 11/14 + 6/14] = 1

pi(0) * [14/14 + 11/14 + 6/14] = 1

pi(0) * 31/14 = 1

pi(0) = 14/31

pi(1) = (11/14) * (14/31) = 11/31

pi(2) = (3/7) * (14/31) = 6/31

Check: 14/31 + 11/31 + 6/31 = 31/31 = 1 (OK)

Q2 Solution

a) Rate matrix Q for states {0,1,2,3}:

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States: lam_n = lambda (all n), mu_n = n*mu

          0            1            2            3
 0  [ -lambda     lambda        0            0     ]
 1  [    mu      -(lambda+mu)  lambda        0     ]
 2  [    0          2*mu     -(lambda+2mu) lambda  ]
 3  [    0           0           3*mu    -(lambda+3mu) ]
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Each row sums to 0 by construction. Off-diagonal entries are non-negative (they are rates). The chain is a birth-death process because only neighbouring states communicate directly.

b) Stationary distribution via detailed balance:

Detailed balance: pi(n) * lam_n = pi(n+1) * mu_{n+1}

pi(n) * lambda = pi(n+1) * (n+1)*mu

pi(n+1) = pi(n) * lambda / ((n+1)*mu)

Apply repeatedly starting from pi(0):

pi(1) = pi(0) * lambda / (1*mu) = pi(0) * (lambda/mu)^1 / 1!

pi(2) = pi(1) * lambda / (2*mu) = pi(0) * (lambda/mu)^2 / 2!

pi(n) = pi(0) * (lambda/mu)^n / n!

Normalise (let r = lambda/mu):

  sum_{n=0}^{inf} pi(0) * r^n / n! = 1

  pi(0) * e^r = 1   [using e^r = sum r^n/n!]

  pi(0) = e^(-r) = e^(-lambda/mu)

Final answer: pi(n) = e^(-lambda/mu) * (lambda/mu)^n / n!

This is a Poisson(lambda/mu) distribution -- the stationary
distribution is Poisson with mean rho = lambda/mu.

c) Expected population E[N]:

Let r = lambda/mu. With pi(n) = e^(-r) * r^n / n!:

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E[N] = sum_{n=0}^{inf} n * e^(-r) * r^n / n!
     = e^(-r) * sum_{n=1}^{inf} n * r^n / n!
     = e^(-r) * sum_{n=1}^{inf} r^n / (n-1)!
     = e^(-r) * r * sum_{n=1}^{inf} r^(n-1) / (n-1)!
     = e^(-r) * r * sum_{k=0}^{inf} r^k / k!    [k = n-1]
     = e^(-r) * r * e^r
     = r

E[N] = lambda / mu
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This is the well-known result: for a Poisson(r) distribution,
the mean equals the parameter r = lambda/mu.

Q3 Solution

a) Testing detailed balance between A and B:

Detailed balance would require: pi(A)*Q(A,B) = pi(B)*Q(B,A)

  pi(A) * 3 = pi(B) * 2

  pi(B)/pi(A) = 3/2

So IF detailed balance holds, we need pi(B) = (3/2)*pi(A).
We will verify this against the solved stationary distribution in (c).

b) Global balance equations (pi*Q = 0):

(Rate leaving state i) = (Rate flowing into state i)

State A: 5*pi(A) = 2*pi(B) + 1*pi(C) ... (I)
State B: 6*pi(B) = 3*pi(A) + 2*pi(C) ... (II)
State C: 3*pi(C) = 2*pi(A) + 4*pi(B) ... (III)

Note: only 2 of the 3 equations are independent (the third follows from the other two), so we use (I), (II), and normalisation.

c) Solving for pi:

From (I): 5*pi(A) = 2*pi(B) + pi(C) => pi(C) = 5*pi(A) - 2*pi(B) ... (i)

Substitute (i) into (II):
6*pi(B) = 3*pi(A) + 2*(5*pi(A) - 2*pi(B))
6*pi(B) = 3*pi(A) + 10*pi(A) - 4*pi(B)
10*pi(B) = 13*pi(A)
pi(B) = 13/10 * pi(A) ... (ii)

From (i): pi(C) = 5*pi(A) - 2*(13/10)*pi(A)
= 5*pi(A) - 13/5 * pi(A)
= (25/5 - 13/5)*pi(A)
= 12/5 * pi(A) ... (iii)

Normalisation: pi(A) + pi(B) + pi(C) = 1
pi(A) * [1 + 13/10 + 12/5] = 1
pi(A) * [10/10 + 13/10 + 24/10] = 1
pi(A) * 47/10 = 1
pi(A) = 10/47

pi(B) = (13/10) * (10/47) = 13/47
pi(C) = (12/5) * (10/47) = 24/47

Check: 10/47 + 13/47 + 24/47 = 47/47 = 1 (OK)

Now test detailed balance between A and B:

pi(B)/pi(A) = 13/10 = 1.3, but we need 3/2 = 1.5 for DB to hold.

13/10 != 3/2, so detailed balance does NOT hold between A and B.

The chain is NOT reversible.